Interim Progress Report

on

A Survey of Deep Learning-Based Reconstruction Methods for Quantum Imaging Systems

Submitted by

Oishik Kar

23BCS089 B.Tech, Computer Science and Engineering

Indian Institute of Information Technology Dharwad

23bcs089@iiitdwd.ac.in

Under the guidance of

Dr. Aswath Babu H.

Assistant Professor, IIIT Dharwad

Department of Computer Science and Engineering

Indian Institute of Information Technology Dharwad

PARIMANA Summer Internship Program 2026

May 2026

Contents

1 Introduction

This document is an interim progress report for the project titled *"A Survey of Deep Learning-Based Reconstruction Methods for Quantum Imaging Systems,"* being carried out under the PARIMANA Summer Internship Program 2026 at Qmet Tech Foundation. The report is intended for submission to both the supervising faculty at IIIT Dharwad and the internship coordination team at Qmet Tech Foundation. It provides a detailed account of the work completed so far, describes the technical approach adopted, summarises the principal findings from the literature review and software development activities, and outlines the current status of the survey manuscript. The overarching goal of the project is to produce a structured, publication-quality survey of deep learning methods that have been applied to image reconstruction in quantum optical systems. The survey covers three principal imaging modalities, namely quantum ghost imaging, compressed sensing with entangled photons, and sub-shot-noise imaging, and additionally examines hybrid classical–quantum reconstruction frameworks. The review encompasses both foundational quantum optics literature (Pittman et al., 1995; Erkmen and Shapiro, 2010) and recent deep learning contributions (Barbastathis et al., 2019).

2 Work Completed

2.1 Literature Search and Taxonomy Development

The first phase of the project involved conducting a systematic literature search across IEEE Xplore, arXiv, Google Scholar, and the Optica Publishing Group digital library. Search queries were constructed by pairing quantum imaging terminology ("ghost imaging," "quantum imaging," "entangled photon imaging," "SPDC," "sub-shot-noise") with deep learning terms ("neural network," "CNN," "GAN," "U-Net," "image reconstruction," "compressed sensing"). Papers were included if they applied a deep learning or neural network method to a quantum or quantum-inspired imaging reconstruction task, or if they provided essential foundational theory. This search yielded approximately 45 candidate papers spanning the period from 1995 to 2025. To organise the collected literature, a three-axis taxonomic classification framework was developed. The first axis categorises papers by imaging modality (ghost imaging, quantum compressed sensing, sub-shot-noise, or hybrid). The second axis classifies the deep learning architecture employed (CNN, GAN, U-Net, autoencoder, fully connected, or quantum-inspired). The third axis captures the reconstruction task (compressed reconstruction, denoising, super-resolution, or phase retrieval). This taxonomy has been applied consistently throughout the survey and has been effective in revealing methodological trends and identifying gaps.

2.2 Detailed Review of Ghost Imaging Methods

The most intensive review effort has been directed at deep learning methods for quantum ghost imaging, which constitutes the most active subfield. The review traced the evolution of the field from the foundational two-photon ghost imaging experiment of Pittman et al. (1995), through the classical and computational ghost imaging variants of Bennink et al. (2002) and Shapiro (2008), to the first application of deep learning by Lyu et al. (2017). The Lyu et al. work demonstrated that a fully connected neural network could reconstruct ghost images at a sampling ratio of only 12.5%, compared to the near-100% ratio required by traditional correlation-based methods (Erkmen and Shapiro, 2010). Subsequent works were reviewed in detail, including the convolutional approach of He et al. (2018), which improved generalisation to unseen object categories and validated on both simulated and experimental data; the computational ghost imaging work of Shimobaba et al. (2018); and the end-to-end framework of Wang et al. (2019), which jointly optimised illumination patterns and the reconstruction network to achieve 3–5 dB PSNR improvement. The review also examined GAN-based ghost imaging reconstruction, noting the tension between improved perceptual quality and the risk of hallucination in scientific imaging contexts. A comparative analysis table of methods, architectures, sampling ratios, and key contributions has been compiled and is included in the survey manuscript.

2.3 Review of Compressed Sensing and Other Domains

The compressed sensing domain has been fully reviewed. The foundational compressive ghost imaging work of Katz et al. (2009) was studied in the context of the compressed sensing theory of Donoho (2006) and Candès and Tao (2006), with particular attention to how quantum correlations provide favourable measurement properties for compressed recovery. The review also covers deep unfolding networks, which parameterise iterative compressed sensing algorithms as trainable neural networks, tracing the development from LISTA (Gregor and LeCun, 2010) and ADMM-Net (Yang et al., 2016) through to ghost imaging-specific applications such as SVD-based deep unfolding (Cheng et al., 2022) and MPDU-Net (Zou et al., 2024). A detailed comparison of data-driven versus model-based reconstruction strategies is included. The review of sub-shot-noise quantum imaging has been completed, covering the experimental demonstrations of Brida et al. (2010) and Samantaray et al. (2017), classical denoising baselines (BM3D, non-local means), deep learning denoising architectures (DnCNN, Noise2Noise, CARE networks), and recent work applying deep learning to quantum-correlated imaging systems (Li et al., 2021; Scattarella et al., 2023; Cai et al., 2023). The hybrid classical–quantum architectures section has also been completed, covering tensor network approaches (Stoudenmire and Schwab, 2016), quantum kernel methods (Havlíček et al., 2019), quanvolutional networks (Henderson et al., 2020), quantum convolutional neural networks (Cong et al., 2019), and a detailed discussion of NISQ hardware limitations including the barren plateau problem (McClean et al., 2018).

2.4 Software Framework Development

A substantial software framework has been developed in Python using PyTorch to complement the literature survey with empirical validation of the reviewed methods. The framework consists of the following components. A **computational ghost imaging simulator** implements the forward measurement model as described in the survey. The simulator generates configurable illumination patterns (supporting Gaussian random, binary random, and Hadamard bases), computes bucket detector signals via inner product of the pattern with the object, supports configurable measurement noise, and performs traditional differential correlation-based reconstruction. This simulator enables controlled experimentation with different sampling ratios, noise levels, and pattern types. Three **deep learning reconstruction models** have been implemented, corresponding to the principal architectural approaches identified in the survey. The first is a fully connected reconstruction network modelled after Lyu et al. (2017), which maps bucket detector signals directly to image space through dense layers with batch normalisation and dropout. The second is a convolutional architecture inspired by He et al. (2018), which projects bucket signals into a spatial feature map and then applies transposed convolutional layers for upsampling and refinement. The third is a U-Net variant adapted for ghost imaging (Ronneberger et al., 2015), which uses an encoder–decoder structure with skip connections to preserve fine-grained spatial information during reconstruction. A **training and evaluation pipeline** provides automated training with validation-based model selection, standard image quality metrics (peak signal-to-noise ratio and structural similarity index), and visualisation utilities for side-by-side comparison of ground truth, traditional reconstruction, and deep learning reconstruction. The MNIST handwritten digit dataset is used as the default object set, following the convention of the reviewed literature.

2.5 Preliminary Experimental Results

Preliminary experiments have been conducted using the fully connected network at a sampling ratio of 12.5% (98 measurements for 28×28 images). The deep learning model achieved a validation PSNR of approximately 22 dB after training, compared to substantially lower quality from the traditional correlation method at the same sampling ratio. These preliminary results are consistent with the performance reported in the literature and confirm that the software framework is functioning correctly. A more comprehensive set of experiments, comparing all three architectures across multiple sampling ratios, will be conducted as part of the survey's empirical analysis.

3 Survey Manuscript Status

The survey manuscript has been compiled concurrently with the literature review. All eight sections (introduction, background, ghost imaging, compressed sensing, sub-shot-noise imaging, hybrid architectures, discussion, and conclusion) are substantially complete, totalling approximately eleven pages in two-column format. The bibliography contains 47 verified references spanning the foundational quantum optics literature, the deep learning for computational imaging literature, and the specific intersection of the two. The discussion section identifies key open problems including the absence of standardised benchmarks, scalability limitations, and the question of whether quantum correlations provide a genuine advantage when combined with deep learning. The conclusion includes a structured summary of research gaps and recommendations for the community.

4 Remaining Work

The following tasks remain for the final phase of the internship:

1. **Comparative experiments.** Run the full set of reconstruction experiments comparing the three implemented architectures (fully connected, CNN, U-Net) across sampling ratios of 6.25%, 12.5%, 25%, and 50%, and compile a quantitative comparison table with PSNR and SSIM metrics.

2. **Manuscript refinement.** Incorporate supervisor feedback, add experimental results and figures to the survey, and improve the comparative analysis table in the ghost imaging section.

3. **Final review and submission.** Proofread the manuscript, verify all citations, and prepare the final version for submission.

5 Summary

The project is progressing ahead of schedule. The systematic literature review has covered all four domains in depth: ghost imaging, compressed sensing, sub-shot-noise imaging, and hybrid classical–quantum architectures. A comprehensive software framework has been developed and validated, producing results consistent with the published literature. The survey manuscript is approximately 90% complete, with all sections written and all references integrated. The remaining work will focus on conducting the full set of comparative experiments across all three implemented architectures, refining the manuscript based on supervisor feedback, and preparing the final submission.

References

1. Barbastathis, G., Ozcan, A., and Situ, G. (2019). On the use of deep learning for computational imaging. *Optica*, 6(8), 921–943.

2. Bennink, R. S., Bentley, S. J., and Boyd, R. W. (2002). "Two-photon" coincidence imaging with a classical source. *Physical Review Letters*, 89, 113601.

3. Beer, K. et al. (2020). Training deep quantum neural networks. *Nature Communications*, 11, 808.

4. Brida, G., Genovese, M., and Ruo Berchera, I. (2010). Experimental realization of sub-shot-noise quantum imaging. *Nature Photonics*, 4, 227–230.

5. Cai, M. et al. (2023). Surpassing the standard quantum limit of optical imaging via deep learning. *Chinese Optics Letters*, 21, 082701.

6. Candès, E. J. and Tao, T. (2006). Near-optimal signal recovery from random projections. *IEEE Trans. Inform. Theory*, 52(12), 5406–5425.

7. Cheng, Z. et al. (2022). Deep unfolding for SVD compressed ghost imaging. *Applied Physics B*, 128, 45.

8. Cong, I., Choi, S., and Lukin, M. D. (2019). Quantum convolutional neural networks. *Nature Physics*, 15(12), 1273–1278.

9. Donoho, D. L. (2006). Compressed sensing. *IEEE Trans. Inform. Theory*, 52(4), 1289–1306.

10. Erkmen, B. I. and Shapiro, J. H. (2010). Ghost imaging: from quantum to classical to computational. *Adv. Opt. Photon.*, 2(4), 405–450.

11. Gregor, K. and LeCun, Y. (2010). Learning fast approximations of sparse coding. *Proc. ICML*, 399–406.

12. Havlíček, V. et al. (2019). Supervised learning with quantum-enhanced feature spaces. *Nature*, 567, 209–212.

13. He, Y. et al. (2018). Ghost imaging based on deep learning. *Scientific Reports*, 8, 6469.

14. Henderson, M. et al. (2020). Quanvolutional neural networks. *Quantum Machine Intelligence*, 2(1), 2.

15. Katz, O., Bromberg, Y., and Silberberg, Y. (2009). Compressive ghost imaging. *Applied Physics Letters*, 95, 131110.

16. Killoran, N. et al. (2019). Continuous-variable quantum neural networks. *Physical Review Research*, 1, 033063.

17. Li, Z.-M. et al. (2021). Fast correlated-photon imaging enhanced by deep learning. *Optica*, 8(3), 323–328.

18. Lyu, M. et al. (2017). Deep-learning-based ghost imaging. *Scientific Reports*, 7, 17865.

19. McClean, J. R. et al. (2018). Barren plateaus in quantum neural network training landscapes. *Nature Communications*, 9, 4812.

20. Pittman, T. B. et al. (1995). Optical imaging by means of two-photon quantum entanglement. *Physical Review A*, 52, R3429–R3432.

21. Ronneberger, O., Fischer, P., and Brox, T. (2015). U-Net: Convolutional networks for biomedical image segmentation. *Proc. MICCAI*, 234–241.

22. Samantaray, N. et al. (2017). Realization of the first sub-shot-noise wide field microscope. *Light: Science & Applications*, 6, e17005.

23. Scattarella, F. et al. (2023). Deep learning approach for denoising low-SNR correlation plenoptic images. *Scientific Reports*, 13, 19645.

24. Shapiro, J. H. (2008). Computational ghost imaging. *Physical Review A*, 78, 061802.

25. Shimobaba, T. et al. (2018). Computational ghost imaging using deep learning. *Optics Communications*, 413, 147–151.

26. Stoudenmire, E. M. and Schwab, D. J. (2016). Supervised learning with tensor networks. *Proc. NeurIPS*, 29.

27. Wang, F. et al. (2019). Learning from simulation: an end-to-end deep-learning approach for computational ghost imaging. *Optics Express*, 27(18), 25560–25572.

28. Yang, Y. et al. (2016). Deep ADMM-Net for compressive sensing MRI. *Proc. NeurIPS*, 29.

29. Zou, Q. et al. (2024). Single pixel imaging based on multiple prior deep unfolding network. *IEEE Photonics Journal*, 16(2), 1–8.